

Availability & Troubleshooting in the Cloud





COURSE STRUCTURE

Day 1: Networking Fundamentals I

Day 2: Networking Fundamentals II

Day 3: Networking in the Cloud Era

Day 4: Cloud Network Services & Connectivity

Day 5: Availability & Troubleshooting in the Cloud

AGENDA FOR DAY 5

1. High Availability & Resilience
2. Hybrid Connectivity and Availability
3. Troubleshooting Cloud Networking



HIGH AVAILABILITY & RESILIENCE ON AWS

WHAT HIGH AVAILABILITY MEANS

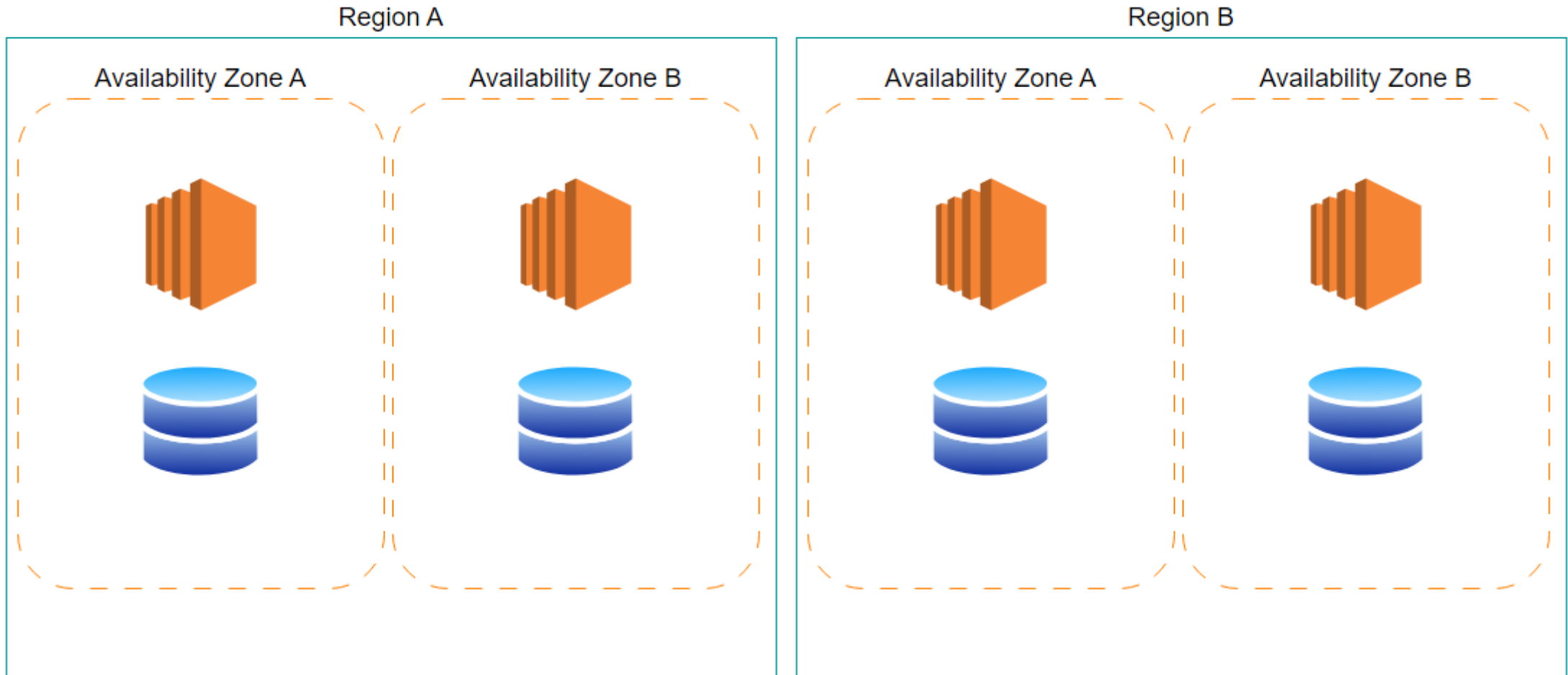


High availability is defined by a system's ability to remain operational and accessible to users for a very high percentage of the time, often 99.9% or higher, with minimal to no downtime.

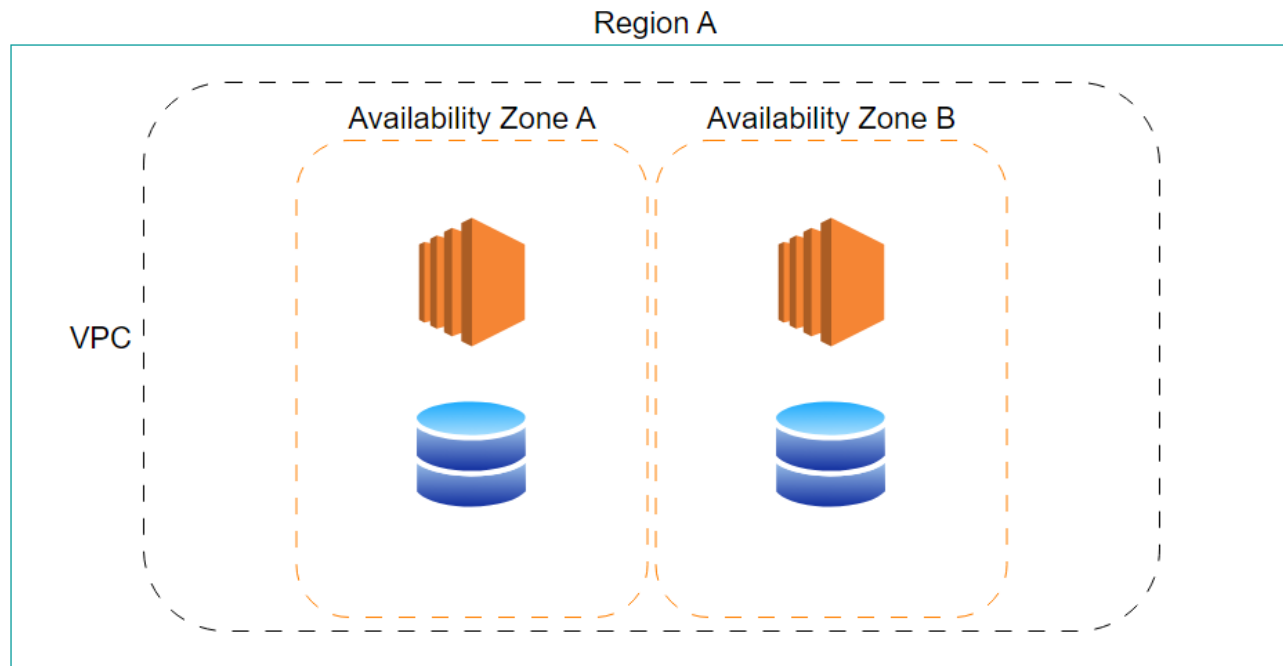
The goal will be to go above 99.99% availability when the systems requires it.

AWS REGION AND AVAILABILITY ZONES

“Design services to remain available even if part of the infrastructure fails.”



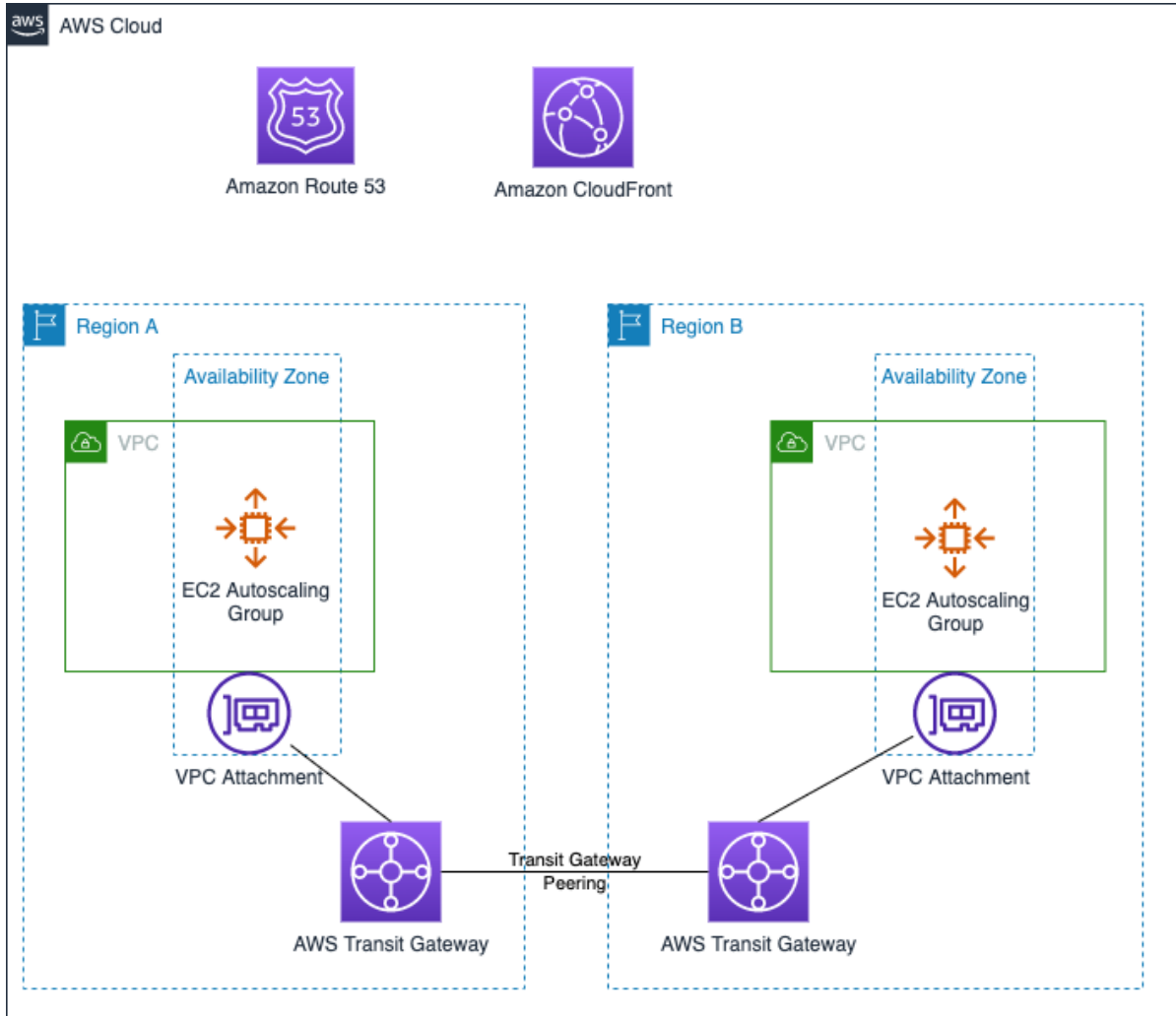
MULTI-AZ ARCHITECTURE



Multi Availability Zone Architecture can be used to achieve redundancy, automatic failover, which leads to increase availability.

Each VPC spans multiple availability zones.

MULTI-REGION ARCHITECTURE



Multi-Region Architecture aims at achieving global load balancing and disaster recovery.

AWS provides tools like Route53 for load balancing and CloudFront as content delivery network service, providing low latency and high transfer speeds.

DESIGNING FOR FAILURE ON AWS

- Assume instances, AZs, and links can fail.
- Automate recovery → Auto Scaling Groups.
- Decouple using SQS, SNS, EventBridge.
- Monitor with CloudWatch + AWS Health Dashboard.
- Store state in durable services (S3, DynamoDB, RDS).

LAB 14: MULTI-AZ WEB APP

- Goal: Build a redundant web tier in two AZs.
- Steps:
 - Create Auto Scaling Group (ASG) across 2 AZs.
 - Attach Application Load Balancer (ALB).
 - Deploy sample app via Launch Template.
 - Stop instances in one AZ → verify traffic shifts.

POP QUIZ:

What does AWS use to achieve fault tolerance within a single region?

- A. Multi-Region replication
- B. Multi-AZ deployments
- C. Elastic IPs
- D. NAT Gateways



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POP QUIZ:

What is the key design philosophy behind AWS high availability?

- A. Assume components will never fail
- B. Assume instances, AZs, and links can fail
- C. Manually restart failed instances
- D. Store state only on local disks



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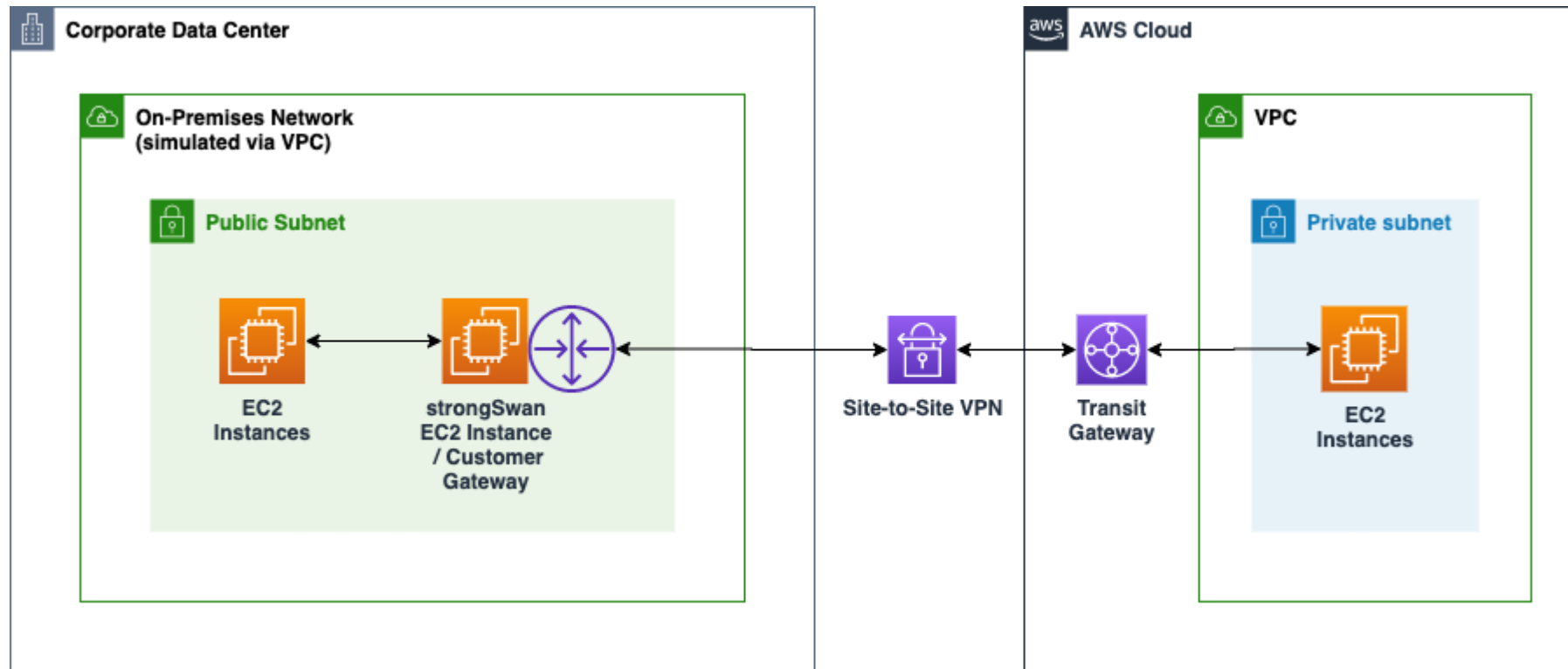
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HYBRID CONNECTIVITY AND AVAILABILITY

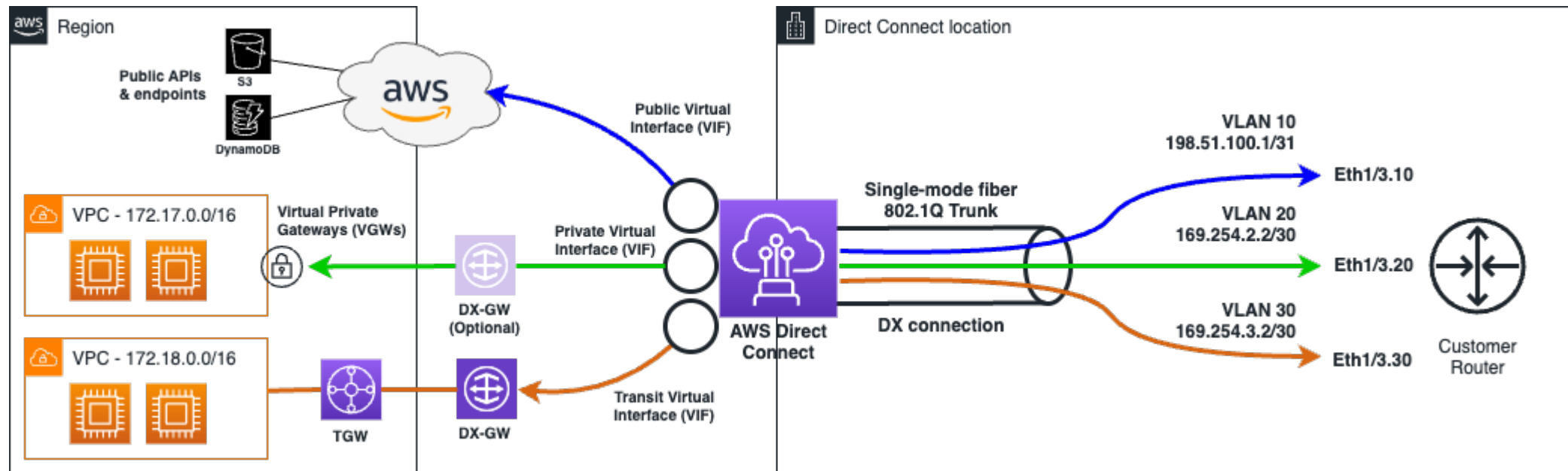
SITE-TO-SITE VPN OVERVIEW

- AWS VPN = VGW (Virtual Private Gateway) + CGW (Customer Gateway).
- Two IPsec tunnels per connection (default redundancy).
- Supports BGP for dynamic failover.



AWS DIRECT CONNECT FOR RESILIENCY

Direct Connect links your internal network to an Direct Connect location over a standard Ethernet fiber-optic cable to AWS Services.



AWS DIRECT CONNECT PRICING

3 factors will determine pricing when it comes to AWS Direct Connect :

- Capacity : Maximum rate that data can be transferred through a network, measured in mega or gigabit per second (gpbs)
- Port hours : Measure the time that a port is provisioned for. Price will depend on the connection type, dedicated or hosted.
- Data Transfer out, or DTO: Refers to the cumulative network traffic that is sent out of AWS through Direct Connect location.

LAB 14: VPN FAILOVER MONITORING

- Goal: Observe AWS VPN failover behavior.
- Steps:
 - Create Site-to-Site VPN with two tunnels.
 - Bring down primary Customer Gateway interface.
 - View CloudWatch metric TunnelState.
 - Confirm secondary tunnel handles traffic.

POP QUIZ:

What is the main benefit of AWS Direct Connect compared to a VPN connection?

- A. Uses the public internet for redundancy
- B. Provides a dedicated, private fiber link for low latency and consistency
- C. Requires less configuration effort
- D. Encrypts traffic by default



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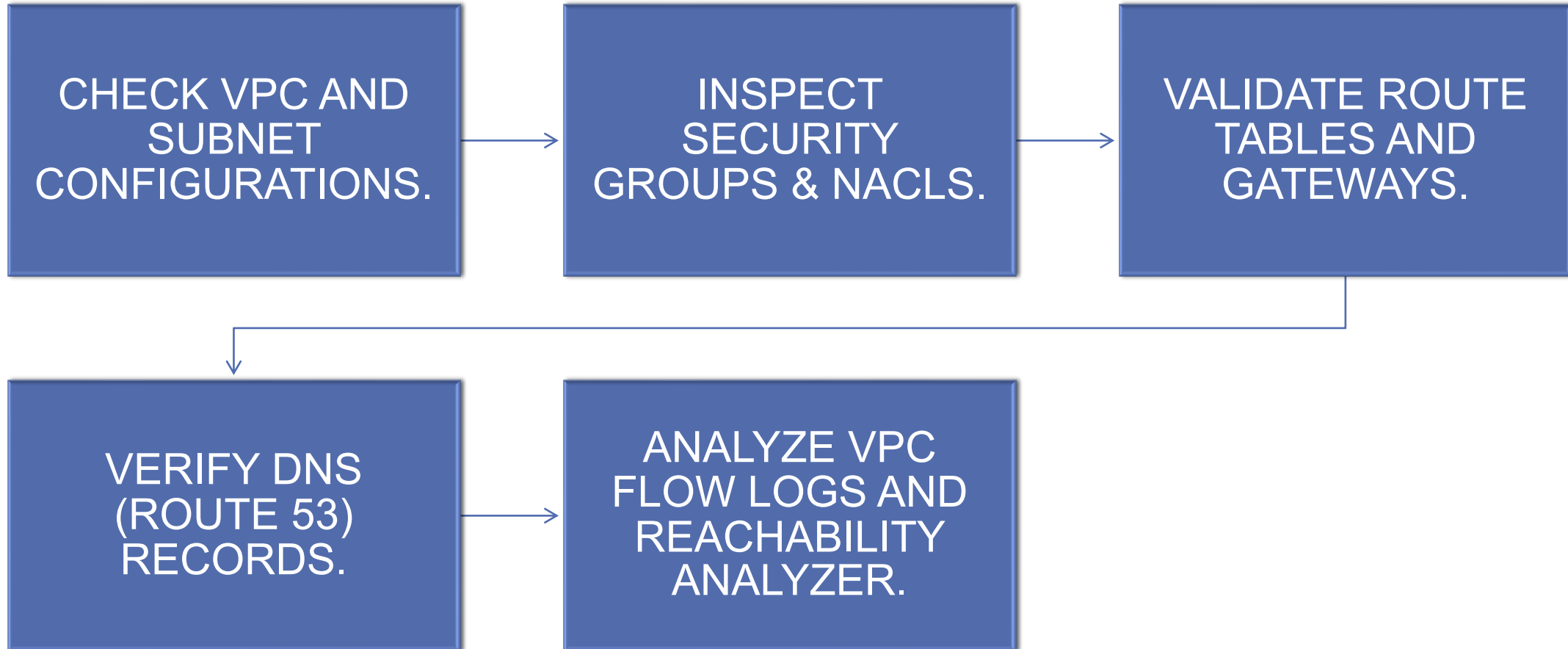
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TROUBLESHOOTING AWS NETWORK CONNECTIVITY

TROUBLESHOOTING WORKFLOW IN AWS

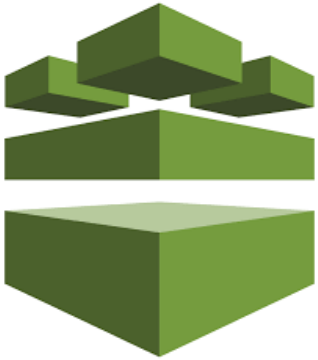


COMMON AWS NETWORK ISSUES

Component	Issue Identified
Security Groups	Missing inbound/outbound rules
NACLs	Overly restrictive stateless filters
Route Tables	No route to Internet Gateway or Peering link
DNS	Private/public mismatch
Load Balancer	Targets failing health checks

AWS NETWORK DIAGNOSTICS TOOLS

AWS Config



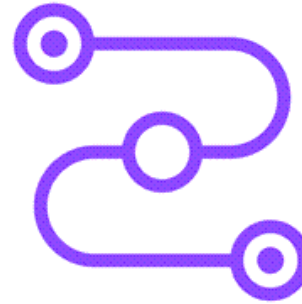
Track Network changes

AWS Cloudwatch



VPC logs and traffic insights

Reachability Analyzer



End-to-end connectivity assessment

CLI Tools



In-instance command line tools

POP QUIZ:

What is the first step in troubleshooting AWS network issues?

- A. Check DNS propagation
- B. Validate VPC, subnets, and routing configurations
- C. Restart EC2 instances
- D. Modify IAM permissions



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LAB 15: TROUBLESHOOTING COMMON VPC ISSUES

- Goal: Diagnose and fix network connectivity problems.
- Steps:
 - Build working environment then intentionally break it.
 - Troubleshoot NAT Gateway connectivity issues.
 - Fix ALB health check and security group problems.
 - Restore VPC peering route configurations.

WRAP-UP : AWS RESILIENCY RECAP

- Multi-AZ and Auto Scaling → continuous availability.
- Route 53 + VPN/Direct Connect → resilient connectivity.
- CloudWatch + SNS → real-time visibility.
- Well-Architected Framework = design checklist.